

MA2031-Hints to solve problems of Assignment I

1. (a) Yes
(b) Yes
(c) No
(d) No
(e) No
(f) No
(g) Yes
(h) No
(i) Yes
(j) No
(k) No
(l) No
(m) No
(n) Yes
2. Yes
3. Yes
4. No
5. (a) No
(b) Yes
(c) Yes
(d) Yes
(e) No
(f) Yes
(g) No
(h) Yes
(i) No
6. $\{0\}$, $\{(x, y) \in \mathbb{R}^2 : y = mx\}$ and $\mathbb{R}^2$ are the subspaces of $\mathbb{R}^2$. $\{0\}$, $\{(x, y, z) \in \mathbb{R}^3 : x/a = y/b = z/c\}$, $\{(x, y, z) \in \mathbb{R}^3 : ax + by + cz = 0\}$ and $\mathbb{R}^3$ are the subspaces of $\mathbb{R}^3$.
7. $b = 0$.
8. Yes.

9. Yes.
10. (a) Yes.
(b) Yes.
(c) Yes.
11. Yes.
12. Trivial.
13. No
14. $\{\alpha_0 + \alpha_1 t^2 + \alpha_2 t^4 + \cdots + \alpha_n t^{2n} : n \in \mathbb{N}\}$.
15. $\mathbb{F}^3$.
16. Follows by observing that every element in S_2 is a linear combination of $u, v_1, v_2, \dots, v_n$ and u is a linear combination of $v_1, v_2, \dots, v_n$
17. Follows from the fact that $L(S)$ is the subspace.
18. It is enough to show that $\{u_1, u_2, \dots, u_n\}$ spans $\{1, t, t^2, t^3, \dots, t^n\}$. Yes, for the second part.
19. Use the fact that $L(S)$ is a subspace containing S and try to use the already known results , Also if $x \in L(S)$ it is in every subspace containing S .
20. No
21. Take one set to be V and try to find another set by removing which element(s) from V so that spanning nature is retained.
22. The given space can be thought of as $\mathbb{R}^2$, so think of a basis for the given space through a basis for $\mathbb{R}^2$
23. Think of the matrices A_{ij} whose ij and ji th entries are 1 and all other entries are zero, and think how many such matrices to be taken to span all the symmetric matrices
24. (a) Follows directly from 17.
(b) Trivial.
(c) Take an element from $L(A \cap B)$, write it as linear combination of elements of $A \cap B$ and observe.
(d) Take two different non-zero vectors such that the linearspan is the same space

- (e) Obviously the R.H.S and L.H.S cannot be equal as one is a subspace and the other is not. Try to find A and B so that the R.H.S is a 1-dimensional subspace and the the linear span of L.H.S is a higher dimensional space
 - (f) Think of two different vectors whose linear span is the same space
25. In $\mathbb{R}^2$, take three different straight lines passing through origin and think what is the sum of two different straight lines passing through origin
26. Let $U \cap W = \{0\}$. Suppose $x = u_1 + w_1 = u_2 + w_2$, show that $u_1 - u_2 \in U \cap W$ and $w_1 - w_2 \in U \cap W$, then the result follows.
27. (a) Take an element $x \in U \cap V + U \cap W$, then $x = a + b$, where $a \in U \cap V$ and $b \in U \cap W$, clearly $x = a + b \in U$.
- (b) Think of 3 different subspaces of $\mathbb{R}^2$, try to make one side is just the zero space and otherside is a higher dimensional space
- (c) If $x \in U + (V \cap W)$, $x = a + b$, $a \in U$ and $b \in (V \cap W)$, now $a + b \in (U + V) \cap (U + W)$.
- (d) Think of 3 different subspaces of $\mathbb{R}^2$ so that one side is $\mathbb{R}^2$ and the other side is a propersubspace of it
